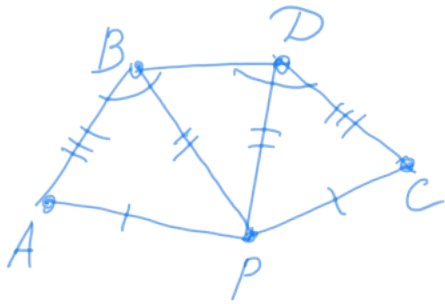


Problem. Let P , A , B , C , and D be points such that $AP = CP$, $BP = DP$ and $AB = CD$. Is it true that $|\angle ABD| = |\angle BDC|$?

The answer is "yes".



By SSS congruence condition,
 $\triangle ABP \cong \triangle CDP$.

Therefore $\angle ABP = \angle PDC$

Since $BP = DP$, we have

$$\angle PBD = \angle BDP.$$

By the axiom on sum of angles,

$$\angle ABD = \angle ABP + \angle PBD = \angle PDC + \angle BDP = \angle BDC.$$

Hence, the statement follows.